6	**(a)**	$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$	1
		$\lambda = \frac{6.63\times10^{-34}}{\sqrt{2\times9.11\times10^{-31}\times7.50\times1.60\times10^{-19}}} = 4.48\times10^{-10}\text{ m}$	1
	(b)	Enough energy to excite atom to -1.43 eV. Hence recoil speeds are 0.25 eV (excited to -1.43 eV), 4.55 eV (excited to -5.73 eV).	1
	(c)	6 transitions shown.	1
	(d)	$\Delta E = \frac{hc}{\lambda}$	1
		$\lambda = \frac{\left(6.63\times10^{-34}\right)\left(3\times10^{8}\right)}{\left(-1.43-(-8.68)\right)\left(1.6\times10^{-19}\right)} = 1.71\times10^{-7}\ m = 171\ nm$	1
		UV region.	1
	(e)	Work function is smaller. Liberated electron not bound to a specific nucleus/atom	1 1